

Here is the complete step-by-step ETL and transformation guide designed specifically for **Power Query Editor inside Power BI Desktop**.

Step 1: Ingest Data & Append Fact Tables

1. Open **Power BI Desktop** → Click **Get Data** → Select **Text/CSV**.
2. Load all 7 CSV files into Power Query by clicking **Transform Data**.
3. **Append Transactions:**
 - Select Transactions_1997 in the Queries pane on the left.
 - Go to **Home Ribbon** → **Combine** → Click **Append Queries as New**.
 - In the dialog box, select **Two tables**:
 - Primary Table: Transactions_1997
 - Second Table: Transactions_1998
 - Rename this new query on the right side (**Query Settings** → **Name**) to **Fact_Transactions**.
4. **Disable Load for Raw Tables (Best Practice):**
 - Right-click Transactions_1997 in the left Queries pane → Uncheck **Enable Load**.
 - Right-click Transactions_1998 → Uncheck **Enable Load**.
 - *(This keeps the staging data in Power Query without loading duplicate records into the Power BI Data Model).*

Step 2: Transform Fact_Transactions

- **Set Data Types:**
 - Select transaction_date and stock_date → Go to **Transform Ribbon** → **Data Type** → Select **Date**.
 - Select product_id, customer_id, store_id, and quantity → Set **Data Type** to **Whole Number**.

Step 3: Transform Customers Dimension

- **Create full_name:**
 - Select first_name, hold Ctrl, and select last_name.
 - Go to **Add Column Ribbon** → Click **Merge Columns**.
 - Set **Separator** to **Space** and enter **New column name** as full_name.
- **Fix Trailing Spaces in upgrade:**

- Select the upgrade column.
- Go to **Transform Ribbon** → **Format** → Click **Trim**.
- **Clean marital_status:**
 - Go to **Add Column Ribbon** → Click **Conditional Column**.
 - Column Name: marital_status_desc.
 - Condition: If marital_status equals "M" then "Married" Else "Single".
- **Extract Birth Year / Age:**
 - Select birthdate → Change **Data Type** to **Date**.
 - Go to **Add Column Ribbon** → **Date** → **Age** (or **Year** → **Year**).
- **Format Postal Code:**
 - Select customer_postal_code → Change **Data Type** to **Text** (*prevents dropping leading zeros*).

Step 4: Transform Products Dimension

- **Handle Null Flags (recyclable & low_fat):**
 - Select recyclable → Go to **Transform Ribbon** → **Replace Values** → Value to Find: null, Replace With: 0.
 - Select low_fat → Go to **Transform Ribbon** → **Replace Values** → Value to Find: null, Replace With: 0.
 - Change Data Type for both columns to **Whole Number**.
- **Set Pricing Precision:**
 - Select product_retail_price and product_cost → Change **Data Type** to **Fixed Decimal Number** (Currency).
- **Calculate Unit Profit Margin:**
 - Go to **Add Column Ribbon** → Click **Custom Column**.
 - Column Name: unit_profit.
 - Formula:

Code snippet

[product_retail_price] - [product_cost]

Step 5: Transform Stores & Regions

- **Set Store Dates:**
 - Select first_opened_date and last_remodel_date → Change **Data Type** to **Date**.

- **Merge Regions into Stores:**

- Select the Stores query.
- Go to **Home Ribbon** → Click **Merge Queries**.
- Select Stores (top) and Regions (bottom).
- Click on region_id in both tables to link them → Join Kind: **Left Outer** → Click **OK**.
- On the newly created Regions column header, click the **Expand Icon** (↔).
- Uncheck region_id and check sales_district and sales_region → Click **OK**.

- **Disable Load on Regions:**

- Right-click the Regions query → Uncheck **Enable Load** (*since its data is now consolidated into Stores*).

Step 6: Transform Returns_1997-1998

- **Set Data Types:**

- return_date → **Date**.
- product_id, store_id, quantity → **Whole Number**.

- Rename query to **Fact_Returns**.

Step 7: Load into Data Model

- Go to **Home Ribbon** → Click **Close & Apply**.

- In Power BI **Model View**, verify the star schema connections:

- Fact_Transactions[product_id] → Products[product_id]
- Fact_Transactions[customer_id] → Customers[customer_id]
- Fact_Transactions[store_id] → Stores[store_id]
- Fact_Returns[product_id] → Products[product_id]
- Fact_Returns[store_id] → Stores[store_id]